

PROJECT APEX

Enterprise LNG Operations Intelligence Platform

VOLUME 2

Data Preparation & Data Modeling

Power Query Cleaning, Validation & Model Building in Power BI Desktop

PART OF THE SERIES

Atlantic LNG Operations Limited (ALOL) — Power BI Project

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Series Overview

Volume 1 — Project Planning & Business Understanding

Volume 2 — Data Preparation & Data Modeling (this document)

Volume 3 — DAX Development

Volume 4 — Dashboard Design & Storytelling

Volume 5 — Portfolio, Documentation & Interview Preparation

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1. Introduction & Objectives of This Volume

Volume 1 established why this project exists and what the dataset contains. Volume 2 is the hands-on, technical volume: it walks through everything required to take the fourteen CSV files already delivered and turn them into a clean, validated, fully-related Power BI semantic model — ready for the DAX measures built in Volume 3 and the dashboards built in Volume 4.

This volume is written to be followed directly inside Power BI Desktop, in the same order a working analyst would actually perform the steps: import, clean, validate, model, then optimize. Every checklist in this document is designed to be used literally — open Power BI Desktop, open this document side by side, and tick items off as you go.

1.1 What This Volume Covers

- A step-by-step import procedure for all 14 CSV files
- A full Power Query cleaning checklist, broken down table by table
- A formal data validation checklist to confirm the data is trustworthy before modeling begins
- Step-by-step relationship building in Model View, including the dual-date challenge on `fact_shutdown_events`
- Guidance on calculated columns vs. measures, and where each belongs
- Performance optimization notes specific to a model of this size (~143,000 rows)

1.2 What This Volume Deliberately Does Not Cover

DAX measure formulas themselves are out of scope here — they belong to Volume 3. Dashboard visuals, colors, and layout belong to Volume 4. This volume stops the moment the model is clean, related, and ready for measures to be written against it.

2. Importing the Dataset into Power BI Desktop

2.1 Before You Start

Confirm you have all 14 CSV files in a single folder on your machine. The recommended folder structure keeps dimension and fact files together but clearly named, exactly as they were delivered:

```
ALOL_Dataset/  dim_calendar.csv  dim_trains.csv  dim_products.csv  dim_countries.csv  
dim_customers.csv  dim_equipment.csv  dim_employees.csv  fact_production.csv  
fact_export_shipments.csv  fact_maintenance.csv  fact_shutdown_events.csv  
fact_hse_incidents.csv  fact_finance.csv  fact_weather.csv
```

2.2 Step-by-Step Import

1. Open Power BI Desktop and select Get Data → Text/CSV.
2. Navigate to your ALOL_Dataset folder and select dim_calendar.csv first — always import dimension tables before fact tables so that Power Query's automatic type detection and relationship suggestions have dimension data to reference.
3. In the preview window, click Transform Data rather than Load — this opens Power Query Editor directly, where all cleaning in Section 3 will take place. Do not load any table directly without passing it through Power Query first, even if it looks clean.
4. Repeat Get Data → Text/CSV for each remaining file. Power BI will queue each one as a separate query in the Queries pane on the left side of Power Query Editor.
5. Once all 14 files are loaded as queries, organize them into two folder groups inside the Queries pane: right-click → New Group, and create a 'Dimensions' group and a 'Facts' group. Drag each query into its appropriate group. This is purely organizational but makes a 14-table model far easier to navigate.
6. Do not click Close & Apply yet — proceed to Section 3 first and apply all cleaning steps per table, then apply once at the end.

2.3 Recommended Query Settings

- Disable 'Enable load' for any staging or reference queries you create later in Volume 3 (none are needed yet in Volume 2)
- Leave 'Include in report refresh' enabled for all 14 tables — this is a static historical dataset, but keeping refresh enabled mirrors real-world practice
- Set Data Type Detection to 'Detect data types based on the first 200 rows' only if your machine struggles with the larger fact tables (fact_export_shipments and fact_finance are the largest); otherwise leave detection on the full dataset for accuracy

3. Power Query Cleaning Checklist

This is the most important section of Volume 2. Each table below has its own checklist of cleaning actions to perform inside Power Query Editor. Work through them in order — dimensions first, then facts — since some fact-table cleaning steps (like confirming foreign key values) are easier to reason about once you have seen the dimension tables they reference.

3.1 General Cleaning Principles Applied to Every Table

- ☐ Promote first row to headers (Power BI usually does this automatically on CSV import — verify it happened correctly)
- ☐ Verify and correct each column's data type explicitly — never leave a column on 'Any' type
- ☐ Rename the query itself to match the table name exactly as used in this documentation (no spaces, consistent casing)
- ☐ Remove any completely blank rows or columns introduced by the CSV export
- ☐ Trim and clean text columns (Transform → Format → Trim, and Format → Clean) to remove leading/trailing whitespace and non-printable characters
- ☐ Check for and remove exact duplicate rows (Home → Remove Rows → Remove Duplicates) applied across the full row, not just the key column
- ☐ Confirm the primary key column has no duplicate values (Right-click column → Remove Duplicates as a temporary check, then undo — never permanently remove rows from a key column without understanding why duplicates exist first)

3.2 dim_calendar — Cleaning Steps

- ☐ Set DateKey to Whole Number
- ☐ Set Date to Date type (not Date/Time — there is no time component in this table)
- ☐ Set Day, DayNumberOfWeek, WeekNumber, MonthNumber, Year, YearMonth, WeekendFlag to Whole Number
- ☐ Set DayName, MonthName, Quarter, FinancialQuarter to Text
- ☐ Verify the date range is continuous with no gaps: sort ascending by Date and confirm 2,192 rows span exactly 1 Jan 2019 to 31 Dec 2024 with no missing calendar days
- ☐ Mark this table as a Date Table in Power BI (Model View → right-click dim_calendar → Mark as Date Table → select Date column) — this enables built-in time intelligence functions in Volume 3

3.3 dim_trains — Cleaning Steps

- ☐ Set TrainID to Whole Number
- ☐ Set CapacityMTPA to Decimal Number
- ☐ Set CommissionDate to Date
- ☐ Set TrainName and Status to Text
- ☐ Confirm exactly 7 rows (Trains 1–7) with no duplicate TrainID values
- ☐ Note that Train 7's Status will read 'Under Construction' — decide now whether Volume 4 dashboards should filter Train 7 out by default, since it has no associated production data (flag this decision in your project notes; it is addressed again in Section 5 of this volume)

3.4 dim_products — Cleaning Steps

- ☐ Set ProductID to Whole Number
- ☐ Set ProductName and ProductCategory to Text
- ☐ Confirm exactly 7 rows with unique ProductID values
- ☐ Confirm ProductCategory only contains the two expected values: 'Gas' and 'Liquid' — use Transform → Group By on ProductCategory as a quick visual check

3.5 dim_countries — Cleaning Steps

- ☐ Set CountryID to Whole Number
- ☐ Set Latitude and Longitude to Decimal Number
- ☐ Set CountryName and Region to Text
- ☐ Confirm exactly 60 rows with unique CountryID values
- ☐ Verify Latitude values fall between -90 and 90, and Longitude values fall between -180 and 180 (Transform → Statistics → Minimum/Maximum on each column) — this confirms the coordinates are valid before they are used in Volume 4's Map Dashboard
- ☐ Confirm Region contains only the expected categories: Europe, Asia, Middle East, Africa, Americas, Oceania

3.6 dim_customers — Cleaning Steps

- ☐ Set CustomerID and CountryID to Whole Number
- ☐ Set CustomerName, CustomerType, ContractType to Text
- ☐ Confirm exactly 30 rows with unique CustomerID values
- ☐ Validate every CountryID value in this table has a matching CountryID in dim_countries (a manual Merge Queries check — left outer join dim_customers to dim_countries on CountryID, then filter

for nulls in the resulting CountryName column; zero null rows confirms referential integrity at the Power Query stage, before the relationship is even built in Model View)

- ☐ Confirm CustomerType only contains: Industrial, Utility, Trader, State
- ☐ Confirm ContractType only contains: Long-term SPA, Short-term, Spot, DES, FOB, CIF

3.7 dim_equipment — Cleaning Steps

- ☐ Set EquipmentID and TrainID to Whole Number
- ☐ Set InstallationDate to Date
- ☐ Set EquipmentCode, EquipmentName, EquipmentType, Area, Manufacturer, Criticality, Status to Text
- ☐ Confirm 335 rows with unique EquipmentID values
- ☐ Validate every TrainID value has a matching TrainID in dim_trains (same Merge Queries technique as Section 3.6)
- ☐ Confirm Criticality only contains: Critical A, Critical B, Major, Minor
- ☐ Confirm Status only contains: Operational, Under Maintenance, Standby

3.8 dim_employees — Cleaning Steps (Special Attention Required)

“This table contains a genuine data quality issue that must be handled deliberately, not silently fixed. 50 rows (EmployeeID 1–50) have a blank Supervisor value. These represent the most senior employees in the simulated company, who — realistically — have no supervisor recorded because they report to executive leadership outside the scope of this dataset. This is documented behavior, not an error to be cleaned away.”

- ☐ Set EmployeeID and YearsExperience to Whole Number
- ☐ Set EmployeeName, Department, Position, Shift, Supervisor, EmploymentType, CertificationLevel to Text
- ☐ Confirm exactly 3,000 rows with unique EmployeeID values
- ☐ Investigate the 50 null Supervisor values (EmployeeID 1–50) — DO NOT delete these rows and DO NOT silently fill them with a fabricated value. The correct treatment is to replace the null with an explicit, descriptive label such as ‘No Supervisor — Senior Role’ using Transform → Replace Values, so that downstream DAX measures and visuals can filter or group on this condition deliberately rather than encountering unexpected blanks
- ☐ Confirm Department contains exactly the 13 expected categories (Production, Maintenance, HSE, Engineering, Finance, HR, IT, Logistics, Commercial, Legal, Operations, Procurement, Quality)
- ☐ Confirm EmploymentType only contains: Permanent, Contract, Secondment, Expatriate

3.9 fact_production — Cleaning Steps

- ☐ Set ProductionID, DateKey, TrainID, ProductID to Whole Number
- ☐ Set GasReceived, LNGProduced, LPGProduced, CondensateProduced, Availability, Utilization, EnergyConsumption, OperatingCost to Decimal Number
- ☐ Confirm 13,152 rows (6 trains × 2,192 days)
- ☐ Validate every DateKey exists in dim_calendar, every TrainID exists in dim_trains, and every ProductID exists in dim_products using the Merge Queries null-check technique
- ☐ Confirm Availability and Utilization values fall between 0 and 100 (or 0 and 1, depending on whether they were stored as percentages or decimals — check this explicitly and standardize to one consistent format, recommended: 0–100 scale stored as Decimal Number, formatted as percentage in the model)
- ☐ Spot-check for negative values in GasReceived, LNGProduced, LPGProduced, CondensateProduced, EnergyConsumption, OperatingCost — none should exist; if found, treat as an import/encoding error and investigate the source row

3.10 fact_export_shipments — Cleaning Steps

- ☐ Set ShipmentID, DateKey, CustomerID, ProductID, DestinationCountryID to Whole Number
- ☐ Set Quantity, UnitPrice, Revenue, FreightCost to Decimal Number
- ☐ Set Vessel, LoadingPort, ShipmentStatus to Text
- ☐ Confirm 35,000 rows with unique ShipmentID values
- ☐ Validate DateKey, CustomerID, ProductID, DestinationCountryID all resolve correctly against their dimension tables
- ☐ Critically validate the calculated relationship: confirm Revenue equals Quantity × UnitPrice for every row (add a temporary custom column = [Quantity] * [UnitPrice] - [Revenue] and confirm it evaluates to 0, or a negligible rounding difference, for all 35,000 rows; remove the temporary column once confirmed)
- ☐ Confirm ShipmentStatus only contains: Delivered, In Transit, Loading, Cancelled
- ☐ Decide and document how Cancelled shipments should be treated in Volume 3's revenue measures — recommended default: exclude Cancelled shipments from Revenue and Quantity totals, but keep them visible in a separate shipment-count measure for operational tracking

3.11 fact_maintenance — Cleaning Steps

- ☐ Set MaintenanceID, DateKey, EquipmentID, EmployeeID to Whole Number
- ☐ Set DowntimeHours, MaintenanceCost to Decimal Number

- ☐ Set MaintenanceType, Status to Text
- ☐ Confirm 25,000 rows with unique MaintenanceID values
- ☐ Validate DateKey, EquipmentID, EmployeeID all resolve correctly against their dimension tables
- ☐ Confirm Status only contains: Completed, In Progress, Deferred, Cancelled
- ☐ Confirm DowntimeHours and MaintenanceCost contain no negative values

3.12 fact_shutdown_events — Cleaning Steps (Special Attention Required)

“This table has two date columns referencing the same calendar dimension. This is the single most technically important modeling decision in the entire project and is addressed fully in Section 5.3 of this volume. The Power Query cleaning steps below are straightforward; the relationship-building complexity comes later, in Model View.”

- ☐ Set ShutdownID, TrainID, StartDateKey, EndDateKey, DurationDays to Whole Number
- ☐ Set LostProduction, ShutdownCost to Decimal Number
- ☐ Set ShutdownType, Reason to Text
- ☐ Confirm 227 rows with unique ShutdownID values
- ☐ Validate that EndDateKey is always greater than or equal to StartDateKey for every row (add a temporary custom column to confirm this logically; investigate any row where it is false)
- ☐ Confirm DurationDays roughly matches the day count between StartDateKey and EndDateKey (allow for the fact that DurationDays was generated independently and may not be an exact calendar subtraction — document any discrepancy rather than silently overwriting one value with the other)
- ☐ Confirm ShutdownType only contains: Planned Turnaround, Emergency Shutdown, Partial Curtailment, Feed Gas Interruption
- ☐ Confirm Planned Turnaround rows have DurationDays mostly between 21 and 35, and Emergency Shutdown rows mostly between 1 and 10, consistent with the original business rule used to generate this data — flag any outliers for review rather than deleting them

3.13 fact_hse_incidents — Cleaning Steps

- ☐ Set IncidentID, DateKey, EmployeeID, EquipmentID, LostTimeHours to Whole Number
- ☐ Set IncidentType, Severity, Area to Text
- ☐ Confirm 5,000 rows with unique IncidentID values
- ☐ Validate DateKey, EmployeeID, EquipmentID all resolve correctly against their dimension tables
- ☐ Confirm Severity only contains: Low, Medium, High, Critical

- ☐ Confirm LostTimeHours is 0 for all Low-severity incidents (consistent with the original business rule) and investigate any exceptions

3.14 fact_finance — Cleaning Steps

- ☐ Set FinanceID, DateKey, ProductID to Whole Number
- ☐ Set Revenue, OPEX, CAPEX, MaintenanceCost, Profit to Decimal Number
- ☐ Confirm 60,000 rows with unique FinanceID values
- ☐ Validate DateKey and ProductID resolve correctly against their dimension tables
- ☐ Critically validate the calculated relationship: confirm Profit equals Revenue – OPEX – CAPEX – MaintenanceCost for every row (same temporary custom-column technique as Section 3.10; remove once confirmed)

3.15 fact_weather — Cleaning Steps

- ☐ Set WeatherID, DateKey to Whole Number
- ☐ Set Temperature, Humidity, Rainfall, WindSpeed to Decimal Number
- ☐ Confirm 2,192 rows — exactly one row per calendar day, with unique DateKey values (this is the one fact table where DateKey should be unique, since it is daily-grain, unlike the other six fact tables)
- ☐ Validate DateKey resolves correctly against dim_calendar
- ☐ Confirm Humidity values fall between 0 and 100, and no negative Rainfall or WindSpeed values exist

4. Data Validation Checklist

Where Section 3 covers cleaning individual tables, this section covers validating the dataset as a whole, once all 14 tables have been cleaned and loaded. This is the formal sign-off checklist before model relationships are built in Section 5. Each item should be explicitly confirmed, not assumed.

4.1 Row Count Validation

Table	Expected Row Count	Validation Method
dim_calendar	2,192	COUNTROWS in a blank measure, or Power Query row count footer
dim_trains	7	Power Query row count footer
dim_products	7	Power Query row count footer
dim_countries	60	Power Query row count footer
dim_customers	30	Power Query row count footer
dim_equipment	335	Power Query row count footer
dim_employees	3,000	Power Query row count footer
fact_production	13,152	Power Query row count footer
fact_export_shipments	35,000	Power Query row count footer
fact_maintenance	25,000	Power Query row count footer
fact_shutdown_events	227	Power Query row count footer
fact_hse_incidents	5,000	Power Query row count footer
fact_finance	60,000	Power Query row count footer
fact_weather	2,192	Power Query row count footer

4.2 Referential Integrity Validation

This is the single most important validation category in this volume. A relationship that looks correct in Model View but silently has orphaned keys will produce dashboards that quietly under-report numbers (Power BI excludes unmatched rows from many visual types by default, without any visible error). Validate every relationship below using the Merge Queries null-check technique described in Section 3.6 before trusting any relationship in Model View.

Relationship Check	Expected Result
Every DateKey in all 7 fact tables exists in dim_calendar	Zero unmatched rows
Every TrainID in fact_production and fact_shutdown_events exists in dim_trains	Zero unmatched rows

Relationship Check	Expected Result
Every ProductID in fact_production, fact_export_shipments, fact_finance exists in dim_products	Zero unmatched rows
Every CustomerID in fact_export_shipments exists in dim_customers	Zero unmatched rows
Every DestinationCountryID in fact_export_shipments exists in dim_countries	Zero unmatched rows
Every CountryID in dim_customers exists in dim_countries	Zero unmatched rows
Every EquipmentID in fact_maintenance and fact_hse_incidents exists in dim_equipment	Zero unmatched rows
Every TrainID in dim_equipment exists in dim_trains	Zero unmatched rows
Every EmployeeID in fact_maintenance and fact_hse_incidents exists in dim_employees	Zero unmatched rows

4.3 Business Rule Validation

These checks confirm the data behaves the way a real LNG operation would, consistent with the business rules defined when the dataset was engineered. Volume 1, Section 2.3 explained why these operational relationships matter — this is where they are formally confirmed in the data itself.

- ☐ Production rows during a known shutdown period (cross-referencing fact_shutdown_events date ranges against fact_production DateKey/TrainID) show visibly reduced Availability — spot-check at least 3 shutdown events manually
- ☐ Corrective maintenance records show, on average, higher MaintenanceCost than Preventive maintenance records — confirm using a quick Group By average in Power Query or a temporary pivot
- ☐ Revenue in fact_export_shipments correlates with Quantity and UnitPrice as already confirmed in Section 3.10
- ☐ Profit in fact_finance is never greater than Revenue (since OPEX, CAPEX, and MaintenanceCost are subtracted and should not be negative) — confirm with a quick filter
- ☐ Planned Turnaround shutdowns cluster in DurationDays between 21–35 and Emergency Shutdowns cluster between 1–10, as already confirmed in Section 3.12

4.4 Data Type Final Confirmation

Before closing Power Query Editor for the final time, open each query's Applied Steps pane and confirm the very last step is a Changed Type step with no 'Any' or auto-detected type left unreviewed. This is a five-minute final pass that prevents subtle DAX errors later (for example, a DateKey accidentally left as Text will silently break every relationship built against dim_calendar).

4.5 Sign-Off Statement

“All 14 tables have been cleaned, all data types explicitly set, all referential integrity checks passed with zero orphaned rows, and all business rule spot-checks confirmed. The dataset is approved to proceed to Model View relationship building.” — use this statement (or your own equivalent) as a personal checkpoint before moving to Section 5. In a real consulting engagement, this would be the point at which sign-off is requested from a data owner or technical lead.

5. Building Relationships in Model View

With all 14 tables cleaned and validated, click Close & Apply in Power Query Editor to load the data into the Power BI data model. This section walks through building every relationship defined in Volume 1, Section 9.3, directly in Model View.

5.1 General Relationship-Building Steps

7. Switch to Model View (the icon on the left-hand navigation rail that looks like three connected boxes).
8. Arrange tables visually with all 7 dimension tables across the top of the canvas and all 7 fact tables below them — this mirrors the conceptual star schema shape and makes the model far easier to read at a glance.
9. For each relationship, either drag the key column from the dimension table directly onto the matching key column in the fact table, or use Home → Manage Relationships → New for more control over cardinality and cross-filter direction.
10. For every relationship, explicitly set Cardinality to One to Many and Cross-filter direction to Single, unless a specific reason (documented below) requires otherwise.
11. Confirm 'Make this relationship active' is checked for the primary relationship on each fact table, and unchecked only for the intentional second relationship on fact_shutdown_events (see Section 5.3).

5.2 Standard Relationships to Build

Build the following 17 relationships first — these are straightforward one-to-many relationships with no special handling required, matching the table in Volume 1 Section 9.3:

#	From	To	Key Column
1	dim_calendar	fact_production	DateKey
2	dim_calendar	fact_export_shipments	DateKey
3	dim_calendar	fact_maintenance	DateKey
4	dim_calendar	fact_hse_incidents	DateKey
5	dim_calendar	fact_finance	DateKey
6	dim_calendar	fact_weather	DateKey
7	dim_trains	fact_production	TrainID
8	dim_products	fact_production	ProductID
9	dim_products	fact_export_shipments	ProductID
10	dim_products	fact_finance	ProductID

#	From	To	Key Column
11	dim_customers	fact_export_shipments	CustomerID
12	dim_countries	fact_export_shipments	DestinationCountryID
13	dim_countries	dim_customers	CountryID
14	dim_equipment	fact_maintenance	EquipmentID
15	dim_equipment	fact_hse_incidents	EquipmentID
16	dim_employees	fact_maintenance	EmployeeID
17	dim_employees	fact_hse_incidents	EmployeeID

5.3 The Dual-Date Challenge — fact_shutdown_events

fact_shutdown_events has two date columns: StartDateKey and EndDateKey, both of which conceptually relate to dim_calendar. Power BI only allows one active relationship between any two tables at a time, so this requires deliberate handling rather than a simple drag-and-drop.

12. Build relationship 18: dim_calendar[DateKey] → fact_shutdown_events[StartDateKey]. Set this as the Active relationship. This means that, by default, any report filtering by date will filter shutdown events based on when they started.
13. Build relationship 19: dim_calendar[DateKey] → fact_shutdown_events[EndDateKey]. Uncheck 'Make this relationship active' — this relationship exists in the model but is inactive by default.
14. Document this decision clearly: the active relationship (start date) is the correct default for most visuals, since stakeholders typically want to know 'when did this shutdown begin' when filtering a timeline. The inactive relationship (end date) will be activated selectively in specific DAX measures using the USERELATIONSHIP() function — for example, a measure answering 'how many shutdowns concluded in this period' — which is covered fully in Volume 3.

"This is one of the most common real-world Power BI modeling patterns, and interviewers in BI/analytics roles frequently ask about it directly. Being able to explain why you chose StartDateKey as active rather than EndDateKey, and how USERELATIONSHIP() solves the inactive case, is strong evidence of genuine modeling competence — this exact explanation is reused in Volume 5's interview preparation guide."

5.4 Relationship Validation After Building

- ☐ Open Manage Relationships and confirm exactly 19 relationships are listed
- ☐ Confirm all relationships show 'One to Many' cardinality with the dimension table on the 'One' side
- ☐ Confirm all relationships show 'Single' cross-filter direction, with no bidirectional relationships present anywhere in the model

- ❑ Confirm exactly one relationship is marked Inactive (the EndDateKey relationship on fact_shutdown_events) and all others are Active
- ❑ In Model View, visually confirm no fact table is connected directly to another fact table — all relationships should flow dimension-to-fact or, in the case of dim_customers/dim_countries, dimension-to-dimension
- ❑ Test the model with a simple visual: drop dim_calendar[Year] and a SUM of fact_production[LNGProduced] onto a table visual and confirm it returns six rows (2019–2024) with non-zero, plausible values — this is the fastest sanity check that the entire model is wired correctly before any DAX is written

6. Calculated Columns vs. Measures — Decisions for This Model

One of the most common mistakes in early-stage Power BI models is overusing calculated columns where a measure would be more efficient, flexible, and performant. This section sets the ground rules that Volume 3 will follow when building the full DAX measure library.

6.1 General Rule Applied in This Project

As a default principle: if a calculation needs to respond to filter context (slicers, visual filters, cross-highlighting), it belongs in a measure. If a calculation is a fixed, row-level attribute that does not change based on how a report is filtered, it can be a calculated column — but even then, a Power Query custom column (computed before the data loads into the model) is almost always preferable to a calculated column (computed after load, consuming more memory in VertiPaq).

6.2 Specific Decisions for This Model

Calculation Need	Decision	Reasoning
Total LNG Production	Measure	Must respond to train/date/product filters
Profit Margin %	Measure	Ratio calculation must respect filter context
Shutdown Duration Category (Short/Medium/Long)	Power Query custom column	Fixed, row-level attribute; computed once before load
Year-over-Year Revenue Growth	Measure with time intelligence	Requires dynamic comparison across filter context
Employee Tenure Band (0–5yr, 6–15yr, 16+yr)	Power Query custom column	Fixed attribute of dim_employees, not filter-dependent
Total Shipment Revenue	Measure	Must aggregate and respond to customer/country/product filters
Incident Severity Weight (numeric score for Low/Medium/High/Critical)	Power Query custom column in fact_hse_incidents	Fixed mapping, used inside later measures, but the mapping itself never changes with filter context

6.3 Note for Volume 3

The two Power Query custom columns identified above (Shutdown Duration Category and Employee Tenure Band) should be added now, while still in Power Query Editor, rather than left for Volume 3. Incident Severity Weight should also be added now since it is a static mapping. Adding these at the modeling stage, rather than as DAX calculated columns later, keeps the model leaner and is considered better practice. Volume 3 will reference these three columns directly when building related DAX measures.

7. Performance Optimization Notes

At roughly 143,000 total rows, this model is not large by enterprise standards, but applying good optimization habits now builds the right discipline and keeps the report responsive as Volume 3's DAX measures and Volume 4's visuals are added.

7.1 Column-Level Optimization

- ☐ Remove any columns not needed for analysis before Close & Apply — for example, if LoadingPort in fact_export_shipments only ever contains one value ('Bonny Island Terminal') across all 35,000 rows, consider removing it from the model and noting the constant value in documentation instead, since a column with no variation adds storage cost with zero analytical value
- ☐ Set Vessel, Reason, and other free-text descriptive columns to 'Do Not Summarize' in their column properties, since Power BI may otherwise default to attempting a Sum or Count summarization on text columns in certain visual types
- ☐ Disable auto date/time hierarchies in Power BI options (File → Options → Current File → Data Load → uncheck 'Auto date/time') — since dim_calendar already provides a proper, explicit date dimension, the auto-generated hidden date tables Power BI creates by default are redundant and waste memory

7.2 Data Type Sizing

Whole Number (Int64) columns consume more memory than necessary for columns with a small value range. Where practical, consider using smaller integer types for ID columns with small ranges (TrainID 1–7, ProductID 1–7) — Power BI's import engine handles this automatically in most cases via VertiPaq compression, but explicitly choosing the smallest appropriate type in Power Query (e.g., using the Decimal Number vs. Fixed Decimal Number distinction for currency fields) avoids unnecessary floating-point storage for monetary values, which should generally use Fixed Decimal Number for precision in financial reporting.

7.3 Query Folding Considerations

Since the source is flat CSV files rather than a database, query folding (where Power Query pushes transformation steps back to the data source) does not apply in the same way it would against SQL Server or a similar relational source. However, the order of operations in Power Query still matters for refresh performance: apply filtering and row-reduction steps (if any) before column transformations and type changes, so that subsequent steps operate on the smallest possible dataset.

7.4 Model Size Check

After Close & Apply, check the model size via File → Options and Settings → Options → Current File, or by examining the file size of the saved .pbix file. A model of this scale should typically save to a .pbix file

under 50–60 MB given VertiPaq's columnar compression — if the file is significantly larger, revisit Section 7.1 and Section 7.2 for columns that may not have compressed well (usually high-cardinality text columns like EmployeeName or Vessel).

8. Volume 2 Summary & Sign-Off

By the end of this volume, the following should be true of your Power BI file:

- ☐ All 14 tables imported, cleaned in Power Query, and loaded into the model
- ☐ Every column has an explicit, correct data type — no columns left on 'Any'
- ☐ dim_calendar is marked as the official Date Table
- ☐ Zero orphaned foreign keys across all 19 relationships, validated via Merge Queries null-checks before relationships were even built
- ☐ 19 relationships built in Model View, all One-to-Many with Single cross-filter direction, with exactly one intentionally inactive relationship (fact_shutdown_events EndDateKey)
- ☐ Three Power Query custom columns added (Shutdown Duration Category, Employee Tenure Band, Incident Severity Weight) in preparation for Volume 3
- ☐ Performance optimization steps applied (auto date/time disabled, unnecessary columns removed, appropriate data types confirmed)

This is the point at which a real consulting engagement would pause for client/stakeholder review before investing further time in DAX development. Take a screenshot of your finished Model View diagram and compare it against the relationship diagram described in Volume 1, Section 10 — they should match exactly.

What Comes Next: Volume 3 Preview

Volume 3 — DAX Development will build the full measure library on top of this now-validated model: core aggregation measures for every fact table, time intelligence measures (year-over-year, period-to-date, rolling averages), the USERELATIONSHIP() measures needed for the fact_shutdown_events dual-date pattern established in Section 5.3, and a complete KPI definition library that Volume 4's dashboards will be built around.

End of Volume 2 — Data Preparation & Data Modeling